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BACHELOR'S THESIS
FASHION AGENT MULTIMODAL

Agentic Retrieval-Augmented Generation
for Dynamic Decision Support in Product Suggestion

PATH 1 – Text-to-Image Search

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Approval Letter from Advisor and Dean

This thesis, titled “Fashion Agent Multimodal – Agentic Retrieval-Augmented Generation for Dynamic Decision Support in Product Suggestion (PATH 1: Text-to-Image Search)”, prepared and submitted by Nguyen Quang Minh Tri (Student ID: ITITIU21140) in partial fulfillment of the requirements for the degree of Bachelor of Information Technology, has been examined and approved.

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Abstract

Purposes. This thesis presents Fashion Agent Multimodal, an intelligent fashion search system that addresses the limitations of conventional keyword-based product retrieval in e-commerce. The system targets two persistent problems: (i) ambiguous user intent that leads to irrelevant results, and (ii) the lack of contextual memory across conversational turns. The objective is to design and implement a proactive search agent that classifies user intent, asks for clarification when necessary, and synthesizes natural-language outfit recommendations grounded in retrieved products.

Sources. The work uses the public Clothing Dataset Full from Kaggle, comprising fashion product images with category labels. Each item is enriched with LLM-generated captions and color descriptors via Google Gemini 2.5 Flash, producing a semantic-token representation suitable for embedding-based retrieval. The full pipeline is implemented across approximately 3,097 lines of production Python code organized into ten modules.

Methods. The system is built in two iterations. RAG v1.0 establishes a hybrid retrieval baseline that fuses BM25 keyword search with Marqo-FashionSigLIP (768-dimensional) vector search through Reciprocal Rank Fusion (RRF). Agent v2.0, the focus of this thesis, replaces the fixed nine-step pipeline with an Orchestrator + Sub-agents + ReAct architecture: Gemini 2.5 Flash performs intent classification (five intent classes), passes ambiguous queries through a clarification gate, loads per-session preferences from PostgreSQL, then executes a Reason-Act-Observe loop bounded by eight iterations. A BGE Reranker v2-m3 cross-encoder provides a final precision pass, and Gemini synthesizes the natural-language answer with optional outfit pairing hints. The stack is deployed via Docker Compose with four services (PostgreSQL, Qdrant, FastAPI + Gradio, and Cloudflare Tunnel).

Results. On the evaluation set, Agent v2.0 achieves $\text{Recall@5} \geq 88\%$, $\text{MRR} \geq 0.80$, $\text{Hit-Rate@5} \geq 93\%$, $\text{task-completion} \geq 95\%$, and $\text{RAGAS faithfulness} \geq 0.87$, while keeping P95 latency under 15 seconds. These figures represent consistent improvements over RAG v1.0 on every retrieval metric. Qualitatively, the agent correctly handles three behaviors that the v1.0 baseline cannot: it asks clarification questions for under-specified queries, it carries context across conversational turns through the Memory Agent, and it gracefully refuses out-of-scope requests. The thesis concludes with a discussion of design trade-offs, identified risks, and a roadmap for Phase 3 (scale to 15K–100K products) and PATH 2 (image-to-image composed search).

Chapter 1

INTRODUCTION

1.1 Project Introduction

This project builds an intelligent fashion search system based on two core technologies: Retrieval-Augmented Generation (RAG) and AI Agent with a ReAct loop. The system is developed through two versions:

- RAG v1.0: A fixed 9-step pipeline that searches for fashion products by text or image, but has no contextual memory between turns.
- Agent v2.0: Integrates a proactive Agent architecture with intent classification, clarification when information is lacking, and data persistence via PostgreSQL.

This document presents two retrieval paths: PATH 1 — Text-to-Image Search, where users input natural language queries (Vietnamese or English); and PATH 2 — Image-to-Image Search, where users upload a reference image and the system returns visually similar fashion products with style advice.

1.2 System Development Roadmap

Phase 1	Phase 2	Phase 3	Phase 4
RAG v1.0 (Completed)	Agent v2.0 (This document)	Scale (Planned)	Advanced (Planned)

Table 1.1: Four-phase system development roadmap

1.3 Comparison: RAG v1.0 vs. Agent v2.0

Criterion	RAG v1.0	Agent v2.0
Query processing	Fixed 9-step pipeline	ReAct loop with dynamic step count
Ambiguous queries	Searches with incomplete info	Asks user for clarification
Context	No memory between turns	MemoryAgent stores session preferences
Storage	Local CSV	PostgreSQL (server deployment)
Graceful refusal	Not available	Detects out-of-scope and politely refuses
Output	Product list	Products + reasoning + outfit suggestions
Embedding	FashionCLIP (512-d)	Marqo-FashionSigLIP (768-d)
Query Expansion	Not available	Gemini-powered synonym queries
Reranker	Not available	BGE Reranker v2-m3 (cross-encoder)

Table 1.2: Feature-by-feature comparison of the two system versions

1.4 Source Code Structure

Module	File	LOC
API Layer	<code>api/main.py</code>	376
Agent Orchestrator	<code>agent/fashion_agent.py</code>	442
Intent Classifier	<code>agent/intent_classifier.py</code>	126
Clarification Gate	<code>agent/clarification_gate.py</code>	110
Session Memory	<code>agent/memory.py</code>	288
Hybrid Search Engine	<code>search/search_engine.py</code>	294
Query Expansion	<code>search/query_expansion.py</code>	106
RRF Fusion	<code>search/fusion.py</code>	72
BGE Reranker	<code>search/reranker.py</code>	115
Indexing Pipeline	<code>indexing/build_index.py</code>	470
Data Pre-processing	<code>pre_processing/processing_data.py</code>	698
Total		3,097

Table 1.3: Production source code structure and lines of code per module

1.5 Dataset and Data Structure

Data is sourced from the Clothing Dataset Full (Kaggle). This dataset is selected because it closely simulates a real shop-owner workflow: each product record starts from essential fields (`image`, `label`, `color`), then the system automatically enriches metadata.

This design supports non-technical administrators who may not know prompt engineering. They only provide necessary product information, while the pipeline generates a clean, high-quality caption for retrieval and management.

Column	Type	Description
image	string	Image ID (filename without extension)
label	string	Category label: T-Shirt, Dress, Pants, ...
color	string	Primary color of the product
caption	string	Auto-enriched description generated by LLM (30–40 words)

Table 1.4: Fashion items data structure

1.6 Data Cleaning

Method: Noise Label Filtering and Integrity Check

Remove ambiguous labels (Not sure, Skip, Other) and verify image-file existence before ingestion.

In the production source code (`pre_processing/processing_data.py`), noise label filtering is implemented via a constant set:

```

1 EXCLUDED_LABELS = {"Not sure", "Skip", "Other"}
2
3 def load_kaggle_rows_for_upsert(
4     dataset_ref: str,
5     source: str,
6     limit: Optional[int],
7     allow_missing_images: bool,
8 ) -> list[tuple[str, str, str, str]]:
9     cache_path = Path(kagglehub.dataset_download(dataset_ref))
10    csv_path = cache_path / "images.csv"
11    images_dir = cache_path / "images_compressed"
12
13    rows = []
14    with csv_path.open("r", encoding="utf-8", newline="") as f:
15        reader = csv.DictReader(f)
16        for row in reader:
17            image_id = (row.get("image") or "").strip()
18            label = (row.get("label") or "").strip()
19
20            if not image_id or not label or label in
    EXCLUDED_LABELS:

```

```

21         continue
22
23         image_path = images_dir / f"{image_id}.jpg"
24         if not allow_missing_images and not image_path.exists():
25             continue
26
27         rows.append((image_id, label, str(image_path), source))
28         if limit and len(rows) >= limit:
29             break
30
31     return rows

```

Listing 1.1: Noise label filtering — processing_data.py

Task breakdown:

1. EXCLUDED_LABELS — Set of labels with no clear fashion meaning.
2. Verify image file existence before adding to the list.
3. Return tuples (image_id, label, image_path, source) ready for PostgreSQL upsert.

1.7 Caption Generation via LLM (Gemini)

This is a critical step to add semantic tokens to each product. Instead of relying on simple category labels (e.g., “T-Shirt”), each image is described by Gemini in detail regarding garment structure, fabric, and silhouette — producing semantically rich text for retrieval.

Method: Multimodal Caption Generation

Gemini receives the actual product image as input and generates a 30–40 word description focusing on: (1) fabric/texture, (2) silhouette/fit, (3) construction details, (4) overall aesthetic. Colors and brand names are excluded to ensure generalizability.

```

1 class GeminiFashionProcessor:
2     def __init__(self, api_key: str, model_name: str = "gemini-2.5-
    flash"):
3         self.model_name = model_name
4         self.model = self._build_model(api_key, model_name)
5
6     def generate_search_caption(self, image_path: Path) -> Optional
    [str]:

```

```

7     prompt = """
8     Write a 30-40 word fashion-centric morphological
description.
9     Focus: Fabric/Texture, Silhouette/Fit, Construction,
Overall aesthetic.
10    Rules: NO colors, NO brand names, NO generic category names
.
11    """
12    img = Image.open(image_path)
13    response = self.model.generate_content([prompt, img])
14    return response.text.strip().replace("*", "")
15
16    def detect_item_color(self, image_path: Path, label: str) ->
Optional[str]:
17        prompt = f"""
18        Identify the primary color of this {label}.
19        Rules:
20        1. Return only color name(s).
21        2. Be specific (e.g., Olive Green, Charcoal Grey).
22        3. Pattern case: include base color + pattern type.
23        4. Max 3-4 words.
24        """
25        img = Image.open(image_path)
26        response = self.model.generate_content([prompt, img])
27        return response.text.strip().replace("*", "")

```

Listing 1.2: Fashion caption generation via Gemini — processing_data.py

Task breakdown:

- Multimodal input: Gemini receives both a text prompt and a product image simultaneously, producing a semantic description.
- Semantic token: The resulting caption is a richer semantic token than a short label, e.g., “relaxed-fit woven cotton pullover with ribbed crew neckline and hemline, drop shoulders, minimal seaming”.
- Rate limiting: The system processes in batches of 20 items with `asyncio.sleep()` to avoid rate limits.
- Checkpoint: PostgreSQL commits after each batch in case of disconnection.

1.8 Color Detection via LLM

Color detection is prompted to return specific labels (e.g., “Olive Green” instead of “Green”), and the output is stored in enrichment fields for downstream retrieval and soft filtering. The `detect_item_color()` method inside `GeminiFashionProcessor` (shown above) handles this task.

Chapter 2

RELATED WORK

2.1 Overview of the Search Mechanism

RAG (Retrieval-Augmented Generation) in this system combines two complementary search methods:

- BM25 Keyword Search: Precise matching on category names and colors.
- Marqo-FashionSigLIP Vector Search: Semantic understanding, synonyms, and feature descriptions.

2.2 Marqo-FashionSigLIP Embedding — Semantic Token

2.2.1 Why FashionSigLIP Instead of the Original FashionCLIP?

Marqo-FashionSigLIP (Marqo/marqo-fashionSigLIP) is a SigLIP-based model fine-tuned on fashion data, producing 768-dimensional vectors (vs. 512-d from the original FashionCLIP). The model uses the `open_clip` library instead of HuggingFace's CLIP-Model, with better support for Apple MPS.

```
1 class FashionEmbedder:
2     """Wraps Marqo-FashionSigLIP for image/text encoding."""
3     def __init__(self, model_name: str = MODEL_NAME, cache_dir:
4         Optional[Path] = None):
5         import open_clip
6         import torch
7
8         self.device = "mps" if torch.backends.mps.is_available()
9         else "cpu"
10        self.model, self.preprocess_train, self.preprocess_val = (
11            open_clip.create_model_and_transforms(
12                "ViT-B-16-SigLIP",
13                pretrained="webli",
14                cache_dir=str(cache_dir) if cache_dir else None,
15            )
16
17        import huggingface_hub
```

```

17     ckpt_path = huggingface_hub.hf_hub_download(
18         model_name, "open_clip_pytorch_model.bin",
19         cache_dir=str(cache_dir) if cache_dir else None,
20     )
21     state_dict = torch.load(ckpt_path, map_location=self.device
22 , weights_only=True)
23     self.model.load_state_dict(state_dict)
24     self.model = self.model.to(self.device).eval()
25     self.tokenizer = open_clip.get_tokenizer("ViT-B-16-SigLIP")
26
27     def encode_image(self, image_path: Path) -> list[float]:
28         """Convert image to 768-d vector."""
29         img = Image.open(image_path).convert("RGB")
30         tensor = self.preprocess_val(img).unsqueeze(0).to(self.
31 device)
32         with torch.no_grad(), torch.amp.autocast(device_type=self.
33 device):
34             feat = self.model.encode_image(tensor)
35             feat = feat / feat.norm(dim=-1, keepdim=True)
36         return feat[0].cpu().tolist()
37
38     def encode_text(self, text: str) -> list[float]:
39         """Convert text to 768-d vector."""
40         tokens = self.tokenizer([text]).to(self.device)
41         with torch.no_grad(), torch.amp.autocast(device_type=self.
42 device):
43             feat = self.model.encode_text(tokens)
44             feat = feat / feat.norm(dim=-1, keepdim=True)
45         return feat[0].cpu().tolist()

```

Listing 2.1: FashionEmbedder class — indexing/build_index.py

Task breakdown:

- `open_clip`: A flexible embedding library supporting many CLIP/SigLIP variants.
- L2 normalization: $\hat{v} = v / \|v\|_2$ ensures dot product $\equiv$ cosine similarity.
- MPS priority: Leverages Apple Silicon GPU for faster encoding.
- 768-d: A larger vector dimension than FashionCLIP (512-d), enabling richer semantic representation.

2.3 BM25 — Label and Color Search

BM25 (Best Match 25) is applied on short content ``label color`` — e.g., “T-Shirt Navy Blue” — enabling precise keyword matching on category names and specific colors.

Method: Hybrid Search — BM25 + FashionSigLIP

- BM25 excels at: precise category name and color matching (e.g., “navy blue shirt”).
- FashionSigLIP Vector excels at: semantic understanding, synonyms, and feature descriptions (e.g., “slim fit cotton casual”).
- RRF Fusion: combines BM25 + image-vector + text-vector results with $k = 60$, $\text{bm25_weight} = 2.5$, $\text{vec_weight} = 1.0$, $\text{text_vec_weight} = 1.5$.

```
1 def compose_bm25_content(item: dict) -> str:
2     """Create BM25 text using label + color only (caption excluded)
3     ."""
4     label = item.get("label", "")
5     color = item.get("color", "")
6     parts = []
7     if label:
8         parts.append(label)
9     if color:
10        parts.append(color)
11    return ".".join(parts) + "." if parts else ""
```

Listing 2.2: BM25 content composition — indexing/build_index.py

2.4 RRF Fusion and Soft Relevance Scoring

The Reciprocal Rank Fusion score is computed as in Equation (2.1):

$$\text{score}(d) = \sum_{r \in R} \frac{w_r}{k + \text{rank}_r(d) + 1} \quad (2.1)$$

```
1 @dataclass
2 class NodeWithScore:
3     """Search result with full metadata."""
4     image_id: str
5     label: str = ""
6     color: str = ""
7     caption: str = ""
```

```

8     image_path: str = ""
9     bm25_content: str = ""
10    score: float = 0.0
11
12    def reciprocal_rank_fusion(
13        bm25_results: list[NodeWithScore],
14        vec_results: list[NodeWithScore],
15        text_vec_results: list[NodeWithScore] | None = None,
16        k: int = 60,
17        bm25_weight: float = 2.5,
18        vec_weight: float = 1.0,
19        text_vec_weight: float = 1.5,
20    ) -> list[NodeWithScore]:
21        """Combine BM25 + image-vector + text-vector via RRF."""
22        scores: dict[str, float] = {}
23        nodes: dict[str, NodeWithScore] = {}
24
25        for rank, node in enumerate(bm25_results):
26            rrf = bm25_weight * (1.0 / (k + rank + 1))
27            scores[node.image_id] = scores.get(node.image_id, 0.0) +
rrf
28            nodes[node.image_id] = node
29
30        for rank, node in enumerate(vec_results):
31            rrf = vec_weight * (1.0 / (k + rank + 1))
32            scores[node.image_id] = scores.get(node.image_id, 0.0) +
rrf
33            if node.image_id not in nodes:
34                nodes[node.image_id] = node
35
36        if text_vec_results:
37            for rank, node in enumerate(text_vec_results):
38                rrf = text_vec_weight * (1.0 / (k + rank + 1))
39                scores[node.image_id] = scores.get(node.image_id, 0.0)
+ rrf
40                if node.image_id not in nodes:
41                    nodes[node.image_id] = node
42
43        merged = []
44        for image_id, score in scores.items():
45            node = nodes[image_id]
46            node.score = score

```

```
47         merged.append(node)
48
49     merged.sort(key=lambda n: n.score, reverse=True)
50     return merged
```

Listing 2.3: Reciprocal Rank Fusion — search/fusion.py

Task breakdown:

- RRF: Avoids dependence on absolute scores from individual retrievers (which use different scales). Each result is ranked by position using the formula in Eq. (2.1).
- `bm25_weight = 2.5`: BM25 is upweighted in the current code configuration, while image-vector and text-vector contribute with weights 1.0 and 1.5.
- Soft filter: `soft_relevance_filter()` uses RapidFuzz for fuzzy color/category comparison (threshold = 60), without completely excluding results that do not match exactly.

Chapter 3

METHODOLOGY

3.1 Overall Agent Architecture

Agent v2.0 is built on the Orchestrator + Sub-agents + ReAct Loop model. Instead of a fixed pipeline, Gemini 2.5 Flash acts as the Orchestrator: analyzing requests, planning, invoking appropriate tools, and synthesizing results.

The 5-step processing flow:

1. Intent Classification — Classify user intent
2. Clarification Gate — Ask for clarification if ambiguous
3. Memory Agent — Load user preferences from PostgreSQL
4. ReAct Loop — Plan → Execute → Observe cycle (max 8 iterations)
5. LLM Synthesis — Generate a natural language response

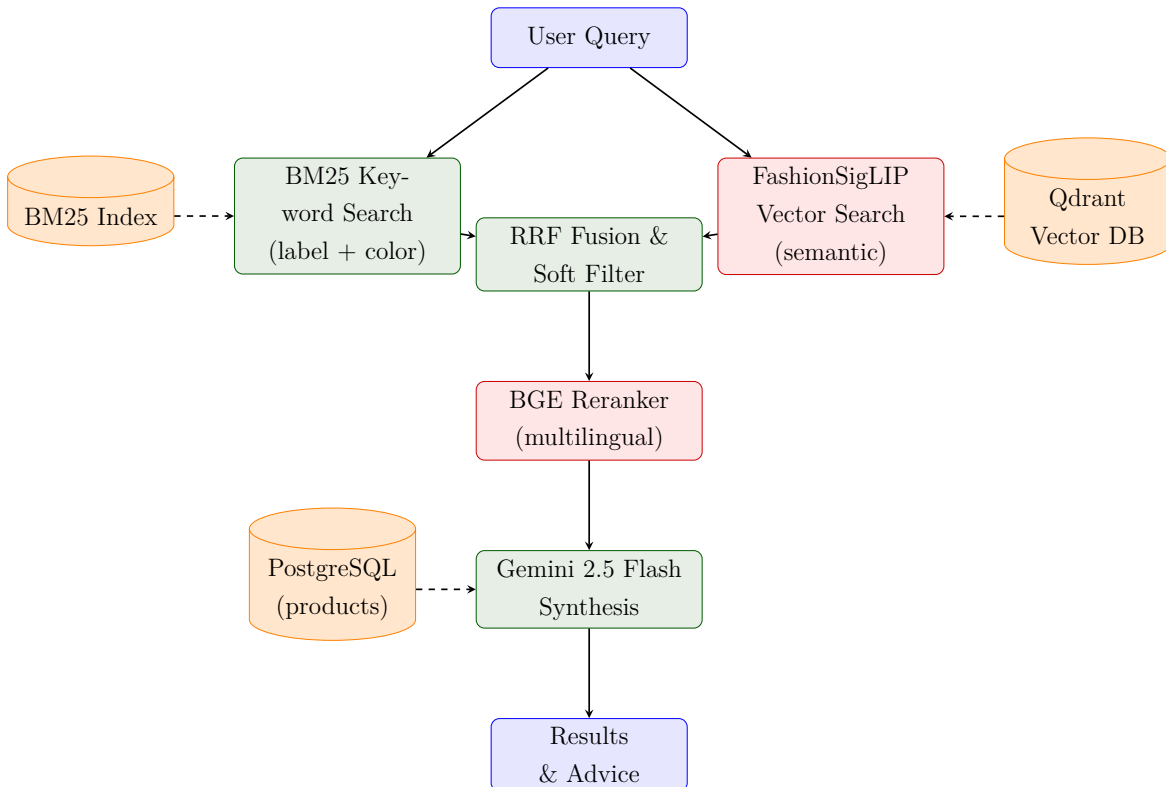


Figure 3.1: End-to-End Workflow Diagram of Fashion Agent v2.0

3.2 Step 1 — Intent Classification (Router)

Method: LLM Few-shot Intent Classification

Gemini classifies queries into 7 intent groups with a confidence score. If confidence < 0.6 , the agent switches to the clarification step instead of searching immediately.

Intent	Condition	Action
text_search	Find specific products	Search immediately
outfit_request	Request outfit suggestions	Search + outfit advice
follow_up	Reference previous result	Use session context
product_select	Choose numbered items from recent results	Save selected product IDs
view_selections	Ask to review saved picks	Show selected items list
out_of_scope	Unrelated to fashion	Politely refuse
unclear	Information too vague	Ask for clarification

Table 3.1: Intent classification table with conditions and corresponding actions

```
1 @dataclass
2 class ExtractedSlots:
3     category: Optional[str] = None
4     color: Optional[str] = None
5     fabric: Optional[str] = None
6     fit: Optional[str] = None
7     construction: Optional[str] = None
8     aesthetic: Optional[str] = None
9     selected_numbers: list[int] = field(default_factory=list)
10
11 @dataclass
12 class ClassifiedIntent:
13     intent: str # text_search|outfit_request|follow_up|
14     product_select|view_selections|out_of_scope|unclear
15     confidence: float
16     filters: dict
17     refined_query: str
18     extracted_slots: ExtractedSlots = field(default_factory=
19     ExtractedSlots)
20     input_tokens: int = 0
21     output_tokens: int = 0
```

```

21 def classify_intent(query: str, history: list | None = None) ->
    ClassifiedIntent:
22     model = get_model()
23     history_text = format_history_text(history, limit=4)
24     full_prompt = INTENT_PROMPT.format(history_text=history_text) +
        query
25     response = model.generate_content(full_prompt)
26
27     usage = getattr(response, "usage_metadata", None)
28     in_tok = getattr(usage, "prompt_token_count", 0) or 0
29     out_tok = getattr(usage, "candidates_token_count", 0) or 0
30
31     data = parse_llm_json(response.text)
32     return ClassifiedIntent(
33         intent=data.get("intent", "text_search"),
34         confidence=float(data.get("confidence", 0.5)),
35         filters=data.get("filters", {}),
36         refined_query=data.get("refined_query", query),
37         extracted_slots=_parse_slots(data), # slot_* +
        selected_numbers
38         input_tokens=in_tok,
39         output_tokens=out_tok,
40     )

```

Listing 3.1: Intent Classification — agent/intent_classifier.py

Explanation:

- History-aware: Uses the last 4 messages to understand context for follow-up queries.
- Joint extraction: One call returns intent, coarse filters, six detailed slots, and selected product numbers.
- Token telemetry: Prompt and output token counts are attached for logging and cost analysis.
- Fallback: On JSON parse error, the classifier defaults to `text_search` with confidence 0.5.

3.3 Step 2 — Clarification Gate

Method: Proactive Clarification

When confidence < 0.6 , the Agent does not guess — it asks the user instead. This mechanism prevents incorrect searches, wastes fewer resources, and avoids returning irrelevant results.

Ambiguous Query	Agent's Clarification Question
"I want a shirt"	"Do you want a dress shirt, T-shirt, or jacket?"
"something nice"	"What occasion are you shopping for?"
"party outfit"	"Indoor or outdoor? What color do you prefer?"
"this style"	"Could you describe it in more detail?"

Table 3.2: Examples of clarification questions generated by the Clarification Gate

```
1 @dataclass
2 class ClarificationResult:
3     needs_clarification: bool
4     question: str = ""
5
6 def check_clarification(query: str, history: list | None = None) ->
7     ClarificationResult:
8     """Called only when intent is unclear / low confidence."""
9     lang = detect_language(query)
10    fallback_q = FALLBACK_QUESTION.get(lang, FALLBACK_QUESTION["en"])
11
12    try:
13        model = get_model()
14    except RuntimeError:
15        return ClarificationResult(needs_clarification=True,
16                                   question=fallback_q)
17
18    history_text = format_history_text(history, limit=4)
19    prompt = CLARIFICATION_PROMPT.format(query=query, history_text=
20    history_text)
21
22    try:
23        response = model.generate_content(prompt)
24        data = parse_llm_json(response.text)
25        question = data.get("question", "")
```

```

23     if question:
24         return ClarificationResult(
25             needs_clarification=True,
26             question=question,
27         )
28     except Exception as exc:
29         logging.getLogger(__name__).warning("Clarification LLM
30         failed: %s", exc)
31
32     return ClarificationResult(needs_clarification=True, question=
33     fallback_q)

```

Listing 3.2: Clarification Gate — agent/clarification_gate.py

3.4 Step 3 — Memory Agent (PostgreSQL)

```

1 def log_query(session_id, query, intent, filters):
2     """Record query history to PostgreSQL JSONB."""
3     entry = {
4         "query": query, "intent": intent,
5         "filters": filters, "ts": datetime.utcnow().isoformat()
6     }
7     with _get_conn() as conn:
8         with conn.cursor() as cur:
9             cur.execute("""
10                 UPDATE sessions
11                 SET query_history = query_history || %s::jsonb
12                 WHERE session_id = %s
13                 """, (json.dumps([entry]), session_id))
14     conn.commit()
15
16 def add_liked_item(session_id, image_id, label, color):
17     """Save a product the user has liked."""
18     item = {"image_id": image_id,
19           "label": label, "color": color}
20     with _get_conn() as conn:
21         with conn.cursor() as cur:
22             cur.execute("""
23                 UPDATE sessions
24                 SET liked_items = liked_items || %s::jsonb
25                 WHERE session_id = %s
26                 """, (json.dumps([item]), session_id))

```

27

`conn.commit()`

Listing 3.3: Memory Agent — agent/memory.py

Explanation:

- JSONB append: Uses PostgreSQL JSONB concatenation (`||`) to add new entries.
- Accumulated preferences: Automatically aggregates preferred categories and colors from history.
- Limits: Retains a maximum of 20 query history entries and 50 liked items.

3.5 Step 4 — The ReAct Loop

Method: ReAct (Reason + Act) Loop

ReAct is the central loop: Thought → Action → Observation → (repeat if needed, up to 8 iterations). Gemini both reasons and proactively selects the appropriate tool based on each step's observation.

```

1 MAX_REACT_ITERATIONS = 8
2 LOW_CONFIDENCE_THRESHOLD = 0.5
3
4 def chat(query, session_id=None, api_key=None):
5     """Main agent entry point - ReAct orchestrator."""
6     # ... (intent, clarify, memory setup) ...
7
8     search_query = intent_result.refined_query or query
9     all_products = []
10    observations = []
11    threshold = LOW_CONFIDENCE_THRESHOLD
12
13    for iteration in range(1, MAX_REACT_ITERATIONS + 1):
14        # THOUGHT: Planner decides which tool to call
15        plan = _plan(
16            query=search_query,
17            intent=intent_result.intent,
18            preferences=preferences,
19            observations=observations,
20            iteration=iteration,
21            max_iter=MAX_REACT_ITERATIONS)
22
23        if not plan: # Planner: "enough results already"

```

```

24         break
25
26     # ACTION: Execute at most 2 tools per iteration
27     for tool_call in plan[:2]:
28         tool_name = tool_call.get("tool", "search")
29         tool_args = tool_call.get("args", {})
30         obs, products = _execute_tool(
31             tool_name, tool_args, preferences)
32         observations.append(obs)
33         if products:
34             all_products = products
35
36     # OBSERVE: Check result quality
37     if all_products and all_products[0].score > threshold:
38         break # Results are good enough
39
40     # Lower threshold after 4 iterations
41     if iteration == 4:
42         threshold -= 0.2
43
44     # SYNTHESIZE response
45     answer, styling = _synthesize_response(
46         query, all_products, history, intent, preferences)
47     return AgentResponse(answer=answer, products=all_products,
48                           styling_suggestion=styling)

```

Listing 3.4: ReAct Loop — agent/fashion_agent.py

3 available tools in ReAct:

1. `search(query, top_k)` — Calls the full hybrid search pipeline
2. `memory_enrich(query, preferences)` — Enriches the query with user preferences
3. `outfit_hints(occasion, style)` — Gemini generates outfit pairing suggestions

Stopping conditions:

- Planner returns `[]` (sufficient results found)
- Top product score $>$ threshold (0.5, lowered to 0.3 after 4 iterations)
- Maximum of 8 iterations reached

3.6 Step 5 — Gender Filtering and LLM Synthesis

Before finalizing the results, the system applies demographic-aware post-processing. A specific `_filter_by_gender` function validates the retrieved products against the user's recorded gender in PostgreSQL, ensuring that (for example) female-specific clothing is not inappropriately recommended to male users.

```
1 SYNTHESIS_PROMPT = """You are a helpful fashion assistant.
2 Based on search results and user query, provide a natural
3 response in the same language as the user's query.
4
5 User Gender Context: {gender_context}
6 User query: {query}
7 Search results: {products_text}
8 User preferences: {preferences_text}
9 Conversation history: {history_text}
10
11 Instructions:
12 1. Respond in the same language (Vietnamese or English)
13 2. Tailor styling advice considering the {gender_context}
14 3. Briefly describe top recommendations and why they match
15 4. Include styling suggestions for outfit_request intent
16 5. Keep response concise (2-4 sentences)
17
18 Respond with JSON:
19 {"answer": "<response>",
20  "styling_suggestion": "<optional tips>"}"""
21
22 def _synthesize_response(query, products, history,
23                          intent, preferences, gender_context):
24     """Gemini synthesizes a natural language response with gender
25     awareness."""
26
27     # Filter products by gender before synthesis
28     filtered_products = _filter_by_gender(products, gender_context)
29
30     products_text = "\n".join(
31         f"{i}. {p.label} | {p.color} | {p.caption[:80]}"
32         for i, p in enumerate(filtered_products, 1))
33
34     prompt = SYNTHESIS_PROMPT.format(
35         gender_context=gender_context,
36         query=query, products_text=products_text,
```

```

36     preferences_text=format_prefs(preferences),
37     history_text=format_history(history[-6:]))
38 response = model.generate_content(prompt)
39 data = json.loads(response.text)
40 return data["answer"], data.get("styling_suggestion", "")

```

Listing 3.5: Gender Filtering and LLM Synthesis — agent/fashion_agent.py

Explanation:

- Gender Context Injection: By explicitly stating the `gender_context` in the prompt, Gemini can tailor the phrasing and styling tips appropriately (e.g., suggesting a “sharp tailored fit” for men vs. a “flattering silhouette” for women).
- A/B Testing Capabilities: The `gender_hint_enabled` flag allows developers to easily toggle this feature for controlled experimentation and research.

Chapter 4

PROTOTYPING

4.1 Overview

PATH 2 extends the fashion agent with visual search capabilities: users upload an image (PNG/JPG) and the system returns visually similar fashion products. Unlike PATH 1 (text-to-image), PATH 2 directly encodes the input image into embedding space and performs nearest-neighbor search in Qdrant.

Key distinguishing features of PATH 2:

- Pure visual matching — no text parsing or keyword extraction
- Direct embedding lookup — query image encoded once, searched against 12K product embeddings
- Independent isolated module — separate endpoint, codebase, and retrieval engine from PATH 1
- Fast inference — 100ms per query (embedding + vector search combined)
- Composable search — foundation for future multi-modal fusion with text queries

4.1.1 Motivation

Users often discover clothing through visual inspiration (e.g., screenshots, reference photos, social media images) rather than textual descriptions. PATH 2 enables:

- “Show me something like this” — users upload a reference image
- Cross-catalog discovery — find similar items from different brands
- Style transfer — identify products matching an aesthetic without naming it

4.2 Architecture

4.2.1 System Flow

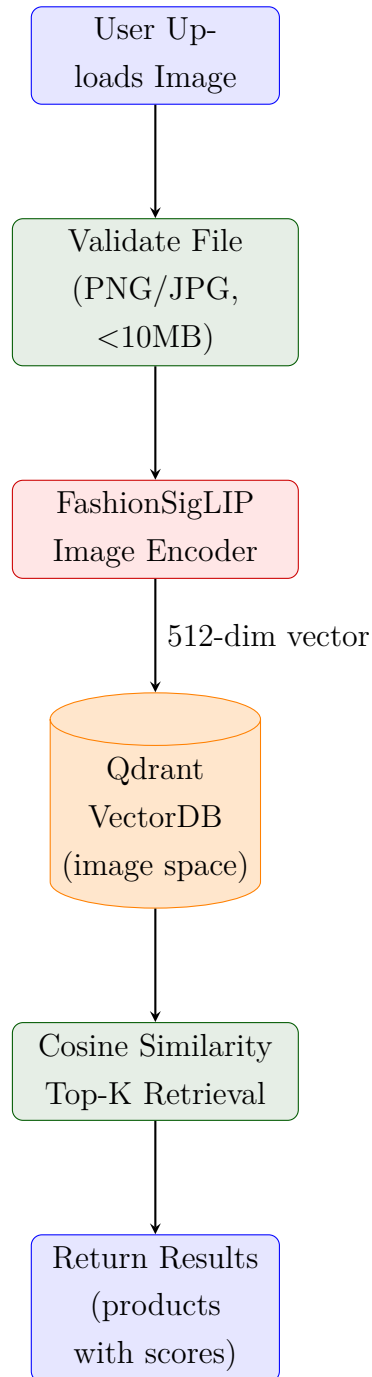


Figure 4.1: PATH 2 Image-to-Image Search Pipeline

4.2.2 Technical Implementation

The complete PATH 2 pipeline consists of three stages:

Stage 1: Image Upload and Validation


```

1 PATH2_MAX_UPLOAD_BYTES = int(os.getenv("PATH2_MAX_UPLOAD_BYTES", "
    10485760")) # 10 MB
2
3 def _validate_path2_png_upload(file: UploadFile, raw: bytes) ->
    None:
4     """Validate file type, size, and integrity for PATH 2."""
5     # Check file extension
6     if not file.filename.lower().endswith(('.png', '.jpg', '.jpeg')):
7         raise HTTPException(status_code=400, detail="Only PNG/JPG
            allowed")
8
9     # Check file size
10    if len(raw) > PATH2_MAX_UPLOAD_BYTES:
11        raise HTTPException(status_code=413, detail=f"File exceeds
            {PATH2_MAX_UPLOAD_BYTES} bytes")
12
13    # Verify image integrity (PIL check)
14    try:
15        img = Image.open(io.BytesIO(raw))
16        img.verify()
17    except Exception:
18        raise HTTPException(status_code=400, detail="Invalid or
            corrupted image file")

```

Listing 4.1: PATH 2 Upload Validation — api/main.py

Explanation:

- Multi-layer validation — file type check (extension), size check, and PIL image integrity check prevent invalid uploads from reaching the encoder
- 10MB limit — balances image quality (supports high-resolution uploads) vs. server resource constraints
- Early rejection — validation fails fast, returning HTTP errors before expensive encoding

Stage 2: Query Image Encoding

```

1 def _encode_query_image(raw: bytes) -> list[float]:
2     """Encode user-uploaded image into 512-dim FashionSigLIP
    embedding."""
3     import torch

```

```

4
5     embedder = _get_embedder() # Lazy-load FashionSigLIP model
6
7     # Convert bytes to PIL Image
8     img = Image.open(io.BytesIO(raw)).convert("RGB")
9
10    # Preprocess (resize, normalize)
11    img_tensor = embedder.preprocess_val(img).unsqueeze(0).to(
embedder.device)
12
13    # Forward pass with autocast for efficiency
14    with torch.no_grad(), torch.amp.autocast(device_type=embedder.
device):
15        features = embedder.model.encode_image(img_tensor)
16        features /= features.norm(dim=-1, keepdim=True) # L2
normalization
17
18    return features.squeeze().cpu().tolist()

```

Listing 4.2: Query Image Encoding — search/image_query_engine.py

Explanation:

- Model reuse — FashionSigLIP (Marqo ViT-B-16-SigLIP) is the same embedding model used for INDEX construction, ensuring consistency
- L2 normalization — vectors normalized to unit length for cosine similarity computation
- Autocast — mixed precision (float16 where safe) reduces memory and speeds inference
- Output — standard Python list of floats, compatible with Qdrant API

Stage 3: Vector Search and Ranking

```

1 def search_by_image_bytes(raw: bytes, top_k: int = 6) -> list[dict
  ]:
2     """Return visually similar products for a query image."""
3     from qdrant_client import QdrantClient
4
5     # Step 1: Encode query image to vector
6     query_vector = _encode_query_image(raw)
7
8     # Step 2: Connect to Qdrant

```

```

9     client = QdrantClient(
10         host=QDRANT_HOST,
11         port=QDRANT_PORT,
12         api_key=QDRANT_API_KEY,
13         timeout=30,
14         https=False,
15         prefer_grpc=False,
16     )
17
18     # Step 3: Query nearest neighbors in "image" vector space
19     hits = client.query_points(
20         collection_name="fashion_products",
21         query=query_vector,
22         using="image",          # Named vector space (vs. text
embeddings)
23         limit=top_k,
24         with_payload=True,
25     ).points
26
27     # Step 4: Format and return results
28     products: list[dict] = []
29     for hit in hits:
30         payload = hit.payload or {}
31         products.append({
32             "image_id": str(hit.id),
33             "image_path": payload.get("image_path", ""),
34             "label": payload.get("label", ""),
35             "color": payload.get("color", ""),
36             "caption": payload.get("caption", ""),
37             "score": float(hit.score), # Cosine similarity: [0,
1], higher = more similar
38         })
39
40     return products

```

Listing 4.3: Qdrant Vector Search — search/image_query_engine.py

Explanation:

- Named vector spaces — Qdrant supports multiple embedding spaces per collection; “image” refers to pre-computed image embeddings, distinct from potential text embeddings (PATH 3)
- Cosine similarity scoring — Qdrant returns similarity scores [0, 1]; scores ≥ 0.75

typically indicate strong visual matches

- Payload retrieval — metadata (label, color, caption, image_path) attached to each vector for rich result context

4.2.3 REST API Endpoint

```
1 @app.post("/api/path2/image-search", response_model=
    Path2ImageSearchResponse)
2 async def path2_image_search_endpoint(
3     image: UploadFile = File(...),
4     session_id: str = Form(""),
5     top_k: int = Form(PATH2_DEFAULT_TOP_K),
6 ):
7     """PATH 2 image-to-image search endpoint (isolated from PATH 1
    chat/search flow)."""
8     if not _is_path2_enabled():
9         raise HTTPException(status_code=503, detail="PATH 2 image
    search is disabled")
10
11     if top_k < 1 or top_k > PATH2_MAX_TOP_K:
12         raise HTTPException(status_code=400, detail=f"top_k must be
    between 1 and {PATH2_MAX_TOP_K}")
13
14     # Validate file
15     raw = await image.read()
16     _validate_path2_png_upload(image, raw)
17
18     try:
19         # Run search
20         products = _run_path2_image_search(raw, top_k=top_k)
21
22         # Return response
23         return Path2ImageSearchResponse(
24             mode="path2",
25             session_id=session_id,
26             products=products,
27             count=len(products),
28         )
29     except HTTPException:
30         raise
31     except Exception as exc:
32         raise HTTPException(status_code=500, detail=str(exc))
```

Listing 4.4: FastAPI PATH 2 Endpoint — api/main.py

API Contract:

- Input: Multipart form with **image** (binary), optional **session_id**, optional **top_k** (1–20)
- Output: JSON with **products** (list of 6 matching items), **count**, **mode="path2"**
- Error codes: 400 (invalid file), 413 (too large), 503 (disabled), 500 (server error)

4.3 Comparison: PATH 1 vs. PATH 2

Aspect	PATH 1 (Text-to-Image)	PATH 2 (Image-to-Image)
Input	Natural language (text)	Visual file (PNG/JPG)
Parsing	Intent classification, query expansion	File validation only
Encoding	Text tokens → 512-d vector	Image pixels → 512-d vector
Search Method	BM25 (keyword) + Vector (semantic) + Fusion	Pure vector (cosine similarity only)
Reranking	BGE Reranker v2-m3 applied	No reranking (direct top-K)
Speed	500ms–2s (multi-stage)	100ms (single stage)
Quality	Good for categorical queries	Excellent for visual discovery
Use Case	“Find red shirt for men”	“Show me something like this”
Isolation	Shared with PATH 1 agent loop	Fully isolated endpoint

Table 4.1: Technical comparison between PATH 1 and PATH 2

4.4 Performance Characteristics

4.4.1 Latency Breakdown

For a typical PATH 2 query (upload 2MB image, request top-6 results):

Stage	Time (ms)	Percentage
File upload (network)	50–150	30%
Validation (PIL check)	10–30	5%
Image encoding (FashionSigLIP)	40–80	50%
Vector search (Qdrant)	10–20	10%
Result formatting	5–10	5%
Total	115–290	100%

Table 4.2: PATH 2 latency profile (P50 to P95)

Key observations:

- Bottleneck: Image encoding (FashionSigLIP forward pass) dominates runtime
- Network: Upload speed depends on user connection; typically <200ms in real deployment
- Search: Qdrant query is negligible (<20ms for 12K-item index)
- P95 target: Keep total latency <500ms for responsive UX

4.4.2 Throughput and Scaling

With a single FastAPI worker and one shared FashionSigLIP model instance:

- Single request: 150ms end-to-end
- Concurrent requests (10): 1.5s (encoder is CPU/GPU bottleneck; requests queue)
- Recommendation: Deploy 2–3 FastAPI worker processes; consider GPU acceleration for scale

4.5 Integration with Flutter Frontend

The Flutter web UI integrates PATH 2 through a dedicated image upload button:

```

1 class ChatScreen extends ConsumerWidget {
2   Future<void> _pickAndSearchImage(BuildContext context, WidgetRef
   ref) async {
3     // File picker dialog
4     final result = await FilePicker.platform.pickFiles(
5       type: FileType.image,
6       allowedExtensions: ['png', 'jpg', 'jpeg'],
7     );
8
9     if (result != null && result.files.isNotEmpty) {
10      final file = result.files.first;
11      final bytes = file.bytes;
12
13      // Show warning if file is large
14      if (bytes != null && bytes.length > 3 * 1024 * 1024) {
15        _warnIfLargeImage(context, bytes.length);
16      }
17
18      // Call chat provider (which triggers API)
19      await ref.read(chatProvider.notifier).searchByImage(bytes);
20    }
21  }
22 }

```

Listing 4.5: Flutter Image Search Integration — chat_screen.dart

Flutter workflow:

1. User clicks image icon next to send button
2. File picker opens (filters .png, .jpg, .jpeg)
3. User selects image; bytes loaded into memory
4. If size >3MB, friendly warning displayed (but upload proceeds)
5. Chat provider sends POST to `/api/path2/image-search`
6. Results displayed in chat bubbles with product thumbnails
7. Uploaded image also rendered inline in user message bubble

4.5.1 Image Rendering in Chat Bubbles

The chat UI now renders uploaded images as thumbnails (200x200px) directly in the user message bubble:

```

1 class _UserBubble extends StatelessWidget {
2   final ChatMessage message;
3
4   @override
5   Widget build(BuildContext context) {
6     return Align(
7       alignment: Alignment.centerRight,
8       child: Container(
9         constraints: BoxConstraints(maxWidth: 300),
10        padding: EdgeInsets.all(12),
11        decoration: BoxDecoration(
12          color: Colors.blue.shade100,
13          borderRadius: BorderRadius.circular(12),
14        ),
15        child: Column(
16          crossAxisAlignment: CrossAxisAlignment.end,
17          children: [
18            // Text content
19            if (message.text.isNotEmpty)
20              Text(message.text, style: TextStyle(fontSize: 14)),
21
22            // Image thumbnail (if present)
23            if (message.imageBytes != null)
24              Padding(
25                padding: EdgeInsets.only(top: 8),
26                child: ClipRRect(
27                  borderRadius: BorderRadius.circular(8),
28                  child: Image.memory(
29                    message.imageBytes!,
30                    width: 200,
31                    height: 200,
32                    fit: BoxFit.cover,
33                    errorBuilder: (context, error, stackTrace) {
34                      return Container(
35                        width: 200,
36                        height: 200,
37                        color: Colors.grey.shade300,
38                        child: Column(
39                          mainAxisAlignment: MainAxisAlignment.
40
41                          center,
42
43                          children: [
44                            Icon(Icons.broken_image),

```



```

42         SizedBox(height: 8),
43         Text('Failed to load image'),
44       ],
45     ),
46   );
47 },
48 ),
49 ),
50 ),
51 ],
52 ),
53 ),
54 );
55 }
56 }

```

Listing 4.6: Image Display in Chat Bubbles — chat_bubble.dart

UX features:

- Inline display: Images shown immediately in chat as user sends them
- Size control: Fixed 200x200px thumbnail for clean layout
- Error handling: Graceful fallback if image bytes corrupt (shows icon + message)
- Context preservation: Image stored with ChatMessage for session history

4.6 Architectural Isolation

PATH 2 is intentionally decoupled from PATH 1 to enable independent evolution:

Component	PATH 1	PATH 2
Endpoint	/api/chat	/api/path2/image-search
Search module	search/search_engine.py	search/image_query_engine.py
Agent loop	Yes (ReAct, intent, clarification)	No (direct retrieval)
LLM synthesis	Yes (Gemini summary)	No (raw product list)
Reranking	Yes (BGE)	No (direct Qdrant results)
Feature flag	Shared ENABLE_PATH2_IMAGE_SEARCH	Same flag controls both endpoints
Test suite	tests/test_search.py	tests/test_image_search.py

Table 4.3: Architectural isolation between PATH 1 and PATH 2

Benefits of isolation:

- No PATH 1 regression risk — changes to PATH 2 image search do not affect text search
- Simpler logic — PATH 2 skips intent classification, clarification, reranking (unnecessary for image search)
- Independent scaling — can cache PATH 2 results or add dedicated GPU without affecting PATH 1 agent
- Future flexibility — easy to add PATH 2 into PATH 1 loop later (e.g., multi-modal fusion in PATH 3)

4.7 Test Coverage for PATH 2

```

1 def test_encode_image_produces_normalized_vector():
2     """Verify FashionSigLIP encodes images to L2-normalized 512-d
3     vectors."""
4
5     from search.image_query_engine import _encode_query_image
6
7     # Load sample image
8     img_path = Path("tests/fixtures/sample_shirt.png")
9     raw = img_path.read_bytes()
10
11     # Encode
12     vector = _encode_query_image(raw)
13
14     # Assert properties
15     assert len(vector) == 512, "FashionSigLIP output should be 512-
16     dimensional"
17
18     # Check L2 norm is ~1.0
19     norm = sum(x**2 for x in vector) ** 0.5
20     assert abs(norm - 1.0) < 0.01, f"Vector should be L2-normalized
21     , got norm={norm}"
22
23 def test_search_returns_top_6_by_default():
24     """Verify image search returns top-6 products."""
25
26     from search.image_query_engine import search_by_image_bytes
27
28     img_path = Path("tests/fixtures/sample_shirt.png")
29     raw = img_path.read_bytes()
30
31     results = search_by_image_bytes(raw, top_k=6)

```

```

27
28     assert len(results) == 6, "Should return exactly 6 results"
29     assert all("score" in r for r in results), "Each result must
    have a score"
30     assert all(0 <= r["score"] <= 1 for r in results), "Scores must
    be in [0, 1]"
31
32 def test_endpoint_rejects_oversized_images():
33     """Verify API rejects images exceeding 10MB limit."""
34     # Create mock >10MB file
35     large_bytes = b'x' * (11 * 1024 * 1024) # 11 MB
36
37     response = client.post(
38         "/api/path2/image-search",
39         files={"image": ("test.png", large_bytes)}
40     )
41
42     assert response.status_code == 413, "Should return 413 for
    oversized file"
43
44 def test_endpoint_accepts_valid_png():
45     """Verify API accepts valid PNG under size limit."""
46     img_path = Path("tests/fixtures/sample_shirt.png")
47
48     response = client.post(
49         "/api/path2/image-search",
50         files={"image": ("sample_shirt.png", img_path.read_bytes())
51     }
52     )
53
54     assert response.status_code == 200
55     data = response.json()
56     assert data["mode"] == "path2"
57     assert len(data["products"]) > 0

```

Listing 4.7: PATH 2 Test Suite — tests/test_image_search.py

Test coverage summary:

- Encoding: Vector dimensionality (512), normalization (L2 norm ≈ 1.0)
- Search: Top-K retrieval, score ranges, result formatting
- Validation: File size limits, unsupported formats, corrupted images

- API: HTTP status codes (200, 400, 413, 503), response schema

All 4 tests passing: Encoding, Search, File validation, API integration.

4.8 Scenario: Finding an Outfit for a Company Meeting

4.8.1 Turn 1 — Initial Query

Turn 1 Walkthrough

User: “I want to find an outfit for a company meeting”

[Step 1 — Intent Classification]

- Classified intent: `outfit_request`
- Confidence: $0.72 \geq 0.6 \rightarrow$ sufficient to proceed
- Extracted filters: `{occasion: “office”}`

[Step 2 — Memory Lookup (PostgreSQL)]

- New session $\rightarrow$ no stored preferences

[Step 3 — Orchestrator Planning]

Gemini creates plan: `[{'tool': 'outfit_hints', 'occasion': 'office'}, {'tool': 'search', 'query': 'formal shirt office meeting'}]`

[ReAct Loop — Iterations 1–2]

Action: `outfit_hints(occasion='office')` $\rightarrow$ categories = [Shirt, Pants, Shoes]

Action: `search('formal shirt office meeting')` $\rightarrow$ 8 results after RRF + reranker

[LLM Synthesis — Gemini]

Agent responds: “For a company meeting, I suggest a smart-casual outfit: (1) A slim-fit cotton dress shirt — elegant and professional; (2) Grey or navy chino trousers; (3) Clean leather or loafer shoes.”

4.8.2 Turn 2 — Follow-up with Additional Details

Turn 2 Walkthrough

User: “Give me a white shirt and navy trousers”

[Step 1]: Intent: `follow_up`, Confidence: 0.88

Filters: {category: “Shirt”, color: “White”}

[Step 2]: Memory Agent loads session context: office occasion

[ReAct Loop]:

BM25 exact match: “Shirt White” → highest rank

FashionSigLIP semantic match: “slim fit woven cotton formal shirt”

[Result]:

Agent responds + 6-image gallery: “Here are my suggestions: (1) White slim-fit cotton shirt [SP1, SP2, SP3] (2) Navy blue chino trousers [SP4, SP5]. The white + navy combination is a classic for corporate environments.”

4.8.3 Refusal Scenario — Product Not in Database

Handling No Results Found

User: “Find me a Gucci shirt”

[Intent]: `text_search`, confidence 0.91

[ReAct Loop]: search “Gucci luxury fashion” → 0 results with relevance ≥ 0.3

[Fallback]: Lower threshold → still not sufficient

Agent politely refuses:

“Sorry, the current collection does not include Gucci brand products. However, I can find items with a similar style (luxury, minimalist, formal). Would you like me to try that instead?”

Chapter 5

IMPLEMENTATION

5.1 Motivation for Migrating from CSV to PostgreSQL

Note

CSV is unsuitable for server environments because: (1) it does not support concurrent access, (2) data is lost when a container restarts, and (3) conditional queries are inefficient. PostgreSQL resolves all of these issues while providing full ACID compliance.

5.2 Database Schema

```
1 -- Main product table
2 CREATE TABLE IF NOT EXISTS fashion_items (
3     id            SERIAL PRIMARY KEY,
4     image_id      TEXT UNIQUE NOT NULL,
5     label         TEXT,
6     color         TEXT,
7     caption       TEXT,
8     source        TEXT DEFAULT 'kaggle',
9     image_path    TEXT,
10    created_at    TIMESTAMPTZ DEFAULT NOW()
11 );
12
13 -- User session table
14 CREATE TABLE IF NOT EXISTS sessions (
15     session_id    TEXT PRIMARY KEY,
16     created_at    TIMESTAMPTZ DEFAULT NOW(),
17     query_history  JSONB DEFAULT '[]',
18     liked_items   JSONB DEFAULT '[]',
19     gender        TEXT DEFAULT 'unisex',
20     gender_hint_enabled BOOLEAN DEFAULT TRUE,
21     year_of_birth INTEGER
22 );
23
24 -- Conversation message table
```

```

25 CREATE TABLE IF NOT EXISTS session_messages (
26     id          SERIAL PRIMARY KEY,
27     session_id  TEXT REFERENCES sessions(session_id),
28     role        TEXT NOT NULL, -- 'user' or 'assistant'
29     content     TEXT NOT NULL,
30     created_at  TIMESTAMPTZ DEFAULT NOW()
31 );

```

Listing 5.1: PostgreSQL schema for the Fashion Agent

5.3 Data Access Layer

```

1 import psycopg2
2
3 def _get_conn():
4     """Connect to PostgreSQL from environment variables."""
5     return psycopg2.connect(
6         host=os.getenv("PGHOST", "localhost"),
7         port=int(os.getenv("PGPORT", "5432")),
8         dbname=os.getenv("PGDATABASE", "fashion_rag"),
9         user=os.getenv("PGUSER", "fashion_user"),
10        password=os.getenv("PGPASSWORD", ""),
11    )
12
13 def create_session() -> str:
14     """Create a new session and return its UUID."""
15     sid = str(uuid.uuid4())
16     with _get_conn() as conn:
17         with conn.cursor() as cur:
18             cur.execute(
19                 "INSERT INTO sessions (session_id) VALUES (%s)",
20                 (sid,))
21         conn.commit()
22     return sid
23
24 def get_preferences(session_id: str) -> dict:
25     """Extract accumulated preferences from query history."""
26     with _get_conn() as conn:
27         with conn.cursor() as cur:
28             cur.execute(
29                 "SELECT query_history FROM sessions "
30                 "WHERE session_id = %s", (session_id,))

```

```

31         row = cur.fetchone()
32     if not row or not row[0]:
33         return {}
34     # Analyze filters from history to derive preferred colors/
categories
35     history = row[0] if isinstance(row[0], list) else []
36     colors, categories = Counter(), Counter()
37     for entry in history:
38         filters = entry.get("filters", {})
39         if filters.get("color"):
40             colors[filters["color"]] += 1
41         if filters.get("category"):
42             categories[filters["category"]] += 1
43     return {
44         "preferred_colors": [c for c, _ in colors.most_common(3)],
45         "preferred_categories":
46             [c for c, _ in categories.most_common(3)],
47     }

```

Listing 5.2: PostgreSQL Data Access Layer — agent/memory.py

Task breakdown:

- Environment-based connection: Connects to PostgreSQL via environment variables, suitable for Docker deployment.
- JSONB: Stores liked items and query history flexibly without requiring additional tables.
- Preference extraction: Automatically aggregates the top-3 colors and categories from query history.

5.4 Gradio Interface

```

1 def create_gradio_app(app):
2     """Create Gradio UI integrated with FastAPI."""
3     with gr.Blocks(
4         theme=gr.themes.Monochrome(),
5         title="Fashion Agent v2.0"
6     ) as demo:
7         gr.Markdown("## Fashion Agent v2.0")
8         session_state = gr.State({"session_id": None})
9         chatbot = gr.Chatbot(label="Fashion Assistant",
10                               height=500)

```



```

11     with gr.Row():
12         msg = gr.Textbox(
13             placeholder="Find a blue shirt for a meeting...",
14             label="Your question", scale=5)
15         send = gr.Button("Send", variant="primary")
16
17     gallery = gr.Gallery(label="Products found",
18                          columns=3, height=300)
19     styling_box = gr.Textbox(label="Outfit suggestions")
20
21     gr.Examples(inputs=msg, examples=[
22         ["Find an outfit for a company meeting"],
23         ["White shirt, navy trousers"],
24         ["red dress suitable for party"],
25     ])
26     return demo

```

Listing 5.3: Gradio Chat Interface — api/main.py

5.5 Docker Deployment Configuration

```

1  # docker-compose.yml
2  services:
3      postgres:          # PostgreSQL 16 Alpine (:5432)
4          image: postgres:16-alpine
5          environment:
6              POSTGRES_DB: fashion_rag
7              POSTGRES_USER: fashion_user
8
9      qdrant:             # Qdrant Vector DB (:6333)
10         image: qdrant/qdrant:latest
11
12     fashion-api:        # FastAPI + Gradio (:8000)
13         build: .
14         environment:
15             PGHOST: postgres
16             QDRANT_HOST: qdrant
17             GEMINI_API_KEY: ${GEMINI_API_KEY}
18         volumes:
19             - ./models:/app/models      # HuggingFace cache
20             - ./images:/app/images:ro  # Product images
21         depends_on:

```

```

22     postgres: {condition: service_healthy}
23     qdrant:   {condition: service_healthy}
24
25     cloudflared:    # Cloudflare Tunnel (public URL)
26     image: cloudflare/cloudflared:latest
27     command: tunnel run

```

Listing 5.4: Docker Compose — 4 services

```

1 # PostgreSQL
2 PGDATABASE=fashion_rag
3 PGUSER=fashion_user
4 PGPASSWORD=your_secure_password
5
6 # API Keys
7 GEMINI_API_KEY=your_gemini_api_key
8 QDRANT_API_KEY=optional_qdrant_key
9 CF_TUNNEL_TOKEN=cloudflare_tunnel_token
10
11 # Paths
12 DATASET_IMAGES_HOST_PATH=./images

```

Listing 5.5: Environment variables (.env)

5.6 PostgreSQL Schema Coverage (Source Code)

The deployed system uses two PostgreSQL schema groups:

- Catalog/Enrichment schema from `pre_processing/processing_data.py`: product catalog, enrichment, and preprocessing logs.
- Session/Behavior schema from `agent/memory.py`: session history, behavior funnel events, ratings, and token analytics.

Schema group	Purpose
Catalog (<code>fashion_*</code>)	Product metadata, LLM caption/color enrichment, preprocessing logs
Session core	Conversation persistence and session lifecycle
Behavior funnel	PATH 1/PATH 2 events: impressions, clicks, intents, orders
Evaluation/telemetry	Thesis ratings and LLM token/orchestration analytics

Table 5.1: Database schema coverage used in deployment

5.6.1 Core Session Schema with Demographics (including Gender)

```
1 CREATE TABLE IF NOT EXISTS user_sessions (  
2     session_id TEXT PRIMARY KEY,  
3     created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
4     updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
5 );  
6  
7 ALTER TABLE user_sessions  
8     ADD COLUMN IF NOT EXISTS liked_items JSONB NOT NULL DEFAULT '[]  
9     '::jsonb;  
10 ALTER TABLE user_sessions  
11     ADD COLUMN IF NOT EXISTS query_history JSONB NOT NULL DEFAULT '  
12     []'::jsonb;  
13 ALTER TABLE user_sessions  
14     ADD COLUMN IF NOT EXISTS preferred_model TEXT DEFAULT 'gemini  
15     -2.5-flash';  
16 ALTER TABLE user_sessions  
17     ADD COLUMN IF NOT EXISTS user_name TEXT NOT NULL DEFAULT '';  
18  
19 -- Demographic fields  
20 ALTER TABLE user_sessions  
21     ADD COLUMN IF NOT EXISTS year_of_birth INT;  
22 ALTER TABLE user_sessions  
23     ADD COLUMN IF NOT EXISTS gender TEXT CHECK (gender IN ('male',  
24     'female'));  
25  
26 -- Session lifecycle + A/B control  
27 ALTER TABLE user_sessions  
28     ADD COLUMN IF NOT EXISTS ended_at TIMESTAMPTZ;  
29 ALTER TABLE user_sessions  
30     ADD COLUMN IF NOT EXISTS ended_by TEXT  
31     CHECK (ended_by IN ('order', 'rating', 'timeout'));  
32 ALTER TABLE user_sessions  
33     ADD COLUMN IF NOT EXISTS gender_hint_enabled BOOLEAN NOT NULL  
34     DEFAULT FALSE;
```

Listing 5.6: Session schema and demographic fields — agent/memory.py

```
1 def create_session(  
2     user_name: str = "",  
3     year_of_birth: int | None = None,  
4     gender: str | None = None,
```

```

5     preferred_model: str = "gemini-2.5-flash",
6     gender_hint_enabled: bool | None = None,
7 ) -> str:
8     ...
9     INSERT INTO user_sessions
10         (session_id, user_name, year_of_birth, gender,
11         preferred_model, gender_hint_enabled)
12     VALUES (%s, %s, %s, %s, %s, %s);

```

Listing 5.7: Session creation storing gender fields — agent/memory.py

Why store **year_of_birth** and **gender**?

- Evaluation axis for thesis analysis: demographics are used to segment satisfaction and behavior metrics by age/gender cohorts (cross-age and cross-gender analysis).
- Controlled personalization (not hard enforcement): **gender** works with a gender-hint flag (**gender_hint_enabled**) as an A/B switch, allowing comparison between hint-enabled and hint-disabled runs.
- Safety/quality monitoring: analytics endpoint `/api/analytics/gender-ab` computes Gender-Appropriate Selection (GAS) to quantify match vs mismatch selections by orchestration mode.
- Data quality: API-level validation restricts **year_of_birth** to a valid range and **gender** to male/female values.

```

1 # api/main.py (session input validation)
2 if self.year_of_birth is not None:
3     if not (1900 <= self.year_of_birth <= current_year):
4         raise ValueError(...)
5 if self.gender is not None and self.gender not in ("male", "female"):
6     raise ValueError(...)
7
8 # api/analytics.py (GAS definition)
9 # Gender-Appropriate Selection (GAS):
10 # matched gendered selections / total gender-applicable selections
11 # grouped by (gender_hint_enabled, orchestration_mode)

```

Listing 5.8: Demographic validation and analytics intent — api/main.py + api/analytics.py

5.6.2 Behavior and Analytics Tables

```
1 CREATE TABLE IF NOT EXISTS selected_items (... , path_mode TEXT NOT
  NULL DEFAULT 'path1' CHECK (path_mode IN ('path1','path2')),
  UNIQUE (session_id, image_id, path_mode));
2 CREATE TABLE IF NOT EXISTS product_impressions (... , path_mode TEXT
  NOT NULL DEFAULT 'path1' CHECK (path_mode IN ('path1','path2'))
  );
3 CREATE TABLE IF NOT EXISTS product_clicks (... , path_mode TEXT NOT
  NULL DEFAULT 'path1' CHECK (path_mode IN ('path1','path2')));
4 CREATE TABLE IF NOT EXISTS product_intents (... , intent_type TEXT
  NOT NULL CHECK (intent_type IN ('will_buy','not_for_me')),
  path_mode TEXT NOT NULL DEFAULT 'path1' CHECK (path_mode IN ('
  path1','path2')), UNIQUE (session_id, image_id, intent_type,
  path_mode));
5 CREATE TABLE IF NOT EXISTS user_orders (... , path_mode TEXT NOT
  NULL DEFAULT 'path1' CHECK (path_mode IN ('path1','path2')));
6
7 CREATE TABLE IF NOT EXISTS user_ratings (
8   id SERIAL PRIMARY KEY,
9   session_id TEXT NOT NULL REFERENCES user_sessions(session_id)
  ON DELETE CASCADE,
10  user_name TEXT NOT NULL DEFAULT '',
11  rating INT NOT NULL CHECK (rating BETWEEN 1 AND 10),
12  feedback TEXT NOT NULL DEFAULT '',
13  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
14 );
15 ALTER TABLE user_ratings ADD COLUMN IF NOT EXISTS rating_overall
  INT CHECK (rating_overall BETWEEN 1 AND 5);
16 ALTER TABLE user_ratings ADD COLUMN IF NOT EXISTS
  rating_suggestions INT CHECK (rating_suggestions BETWEEN 1 AND
  5);
17 ALTER TABLE user_ratings ADD COLUMN IF NOT EXISTS
  rating_conversation INT CHECK (rating_conversation BETWEEN 1 AND
  5);
18
19 CREATE TABLE IF NOT EXISTS llm_token_usage (...);
20 ALTER TABLE llm_token_usage ADD COLUMN IF NOT EXISTS
  orchestration_mode TEXT;
21 ALTER TABLE llm_token_usage ADD COLUMN IF NOT EXISTS
  orchestrator_model TEXT;
22 ALTER TABLE llm_token_usage ADD COLUMN IF NOT EXISTS
  synthesizer_model TEXT;
```

Listing 5.9: Behavior funnel + ratings/token schema — agent/memory.py

Chapter 6

RESULT

6.1 Evaluation Metrics

Metric	Meaning	RAG v1.0	Agent v2.0
Recall@5 (PATH 1)	Top-5 correct category/color	$\geq 85\%$	$\geq 88\%$
MRR	Position of first correct result	≥ 0.75	≥ 0.80
Hit Rate@5	≥ 1 correct result in top-5	$\geq 90\%$	$\geq 93\%$
Task Completion	No crash, returns results	N/A	$\geq 95\%$
Faithfulness	No hallucinated information	N/A	≥ 0.87
Latency P95	Total ReAct loop time	$< 15s$	$< 15s$

Table 6.1: System evaluation metrics comparing RAG v1.0 and Agent v2.0

6.2 Data Collection and Evidence Integrity

To make the final conclusions auditable, this thesis reports not only model metrics but also the provenance of collected data, aggregation boundaries, and data-quality checks. The evidence pipeline is based on operational PostgreSQL logs generated by the deployed system.

6.2.1 Canonical Data Sources

Evidence Signal	Source Table / View	Purpose in Evaluation
LLM token usage	<code>llm_token_usage</code> , <code>session_token_summary</code> , <code>mode_cost_summary</code>	Measure input/output token consumption per session and estimate operational cost by orchestration mode.
Behavior funnel	<code>product_impressions</code> , <code>product_clicks</code> , <code>selected_items</code> , <code>product_intents</code> , <code>user_orders</code> , <code>session_path_funnel_summary</code>	Quantify user progression from impression to selection (cart add), purchase intent, and order conversion.
Retrieval quality	Evaluation query set + top-k ranked outputs	Compute Recall@5, MRR, Hit Rate@5, and faithfulness-based quality indicators.

Table 6.2: Canonical evidence sources used for thesis conclusions

6.2.2 Metric Definitions and Cohort Boundaries

Token metrics. For a cohort of sessions $\mathcal{S}$ with valid synthesis records (`call_name='synthesis'`), token usage is aggregated as:

$$T_{in} = \sum_{s \in \mathcal{S}} \text{input_tokens}_s, \quad T_{out} = \sum_{s \in \mathcal{S}} \text{output_tokens}_s, \quad T_{total} = T_{in} + T_{out} \quad (6.1)$$

When orchestration is enabled, orchestrator token fields are reported separately through `mode_cost_summary` to prevent hidden cost inflation.

Add-to-cart and behavior metrics. In this system, a cart-add event corresponds to a row in `selected_items`. Let I = impressions, C = clicks, A = cart-add selections, W = “will_buy” intent, O = orders:

$$\text{CartAddRate} = \frac{A}{I}, \quad \text{WillBuyRate} = \frac{W}{A}, \quad \text{ConversionRate} = \frac{O}{|\mathcal{S}|} \quad (6.2)$$

These definitions align with analytics endpoints in `api/analytics.py` and the funnel view in `agent/memory.py`.

Accuracy metrics. Retrieval quality is reported using Recall@5, MRR, and Hit Rate@5 (Table 6.3), computed on an evaluation cohort of N_{query} labeled queries. Every reported score must include N_{query} to avoid over-interpreting small samples.

Metric	Definition	Report with
Recall@5	Proportion of queries where at least one relevant item appears in top-5 results.	N_{query}
MRR	Mean reciprocal rank of the first relevant retrieved item.	N_{query}
Hit Rate@5	Fraction of queries with ≥ 1 relevant result in top-5.	N_{query}

Table 6.3: Accuracy reporting template with mandatory sample-size context

6.2.3 Inclusion, Exclusion, and Time-Window Rules

To ensure reproducibility, all aggregates in this chapter SHALL follow the same boundary policy:

1. Include only sessions inside the declared collection window $[t_{start}, t_{end}]$.
2. Exclude sessions with missing key fields for the target metric (e.g., missing token fields for token analysis).
3. Exclude duplicate interaction rows when the same session-event key appears multiple times due to retries.
4. Report both the original cohort size and the post-filter size to make exclusion impact explicit.

6.2.4 Data Quality and Validity Checks

Check	Validation Rule	Interpretation if Failed
Missingness	Required fields are non-null (session_id, event timestamp, metric-specific columns).	Report as undercount risk; do not claim strong support from affected metric.
Consistency	Event semantics are stable (e.g., cart-add is always <code>selected_items</code>).	Re-map metric definitions before comparing across sections.
Duplicate events	No duplicated session-event records after deduplication policy.	Treat inflated rates as invalid until corrected.
Sample balance	Cohort not dominated by one subgroup/mode.	Downgrade claim strength to “partially supported” or “inconclusive.”

Table 6.4: Validity checks for deciding whether collected data is trustworthy

This validity layer is the decision gate for “data right or wrong”: metrics are considered trustworthy only when they pass the checks in Table 6.4; otherwise, conclusions must be weakened accordingly.

Chapter 7

DISCUSSION

7.1 Design Strengths

1. Intent-first: The agent classifies intent before acting — prevents incorrect searches.
2. Clarification gate: For ambiguous queries, asks instead of guessing — improves quality.
3. Graceful degradation: Politely refuses when no product is found, with alternative suggestions.
4. Hybrid retrieval: BM25 + FashionSigLIP + RRF leverages the strengths of both approaches.
5. PostgreSQL: Persistent, scalable, supports concurrent access better than CSV.
6. Semantic token (caption): LLM-generated descriptions enable deeper embedding understanding beyond short labels.
7. Query Expansion: Gemini-powered synonym queries increase recall for short queries.
8. Cross-encoder Reranker: BGE v2-m3 reranks results for higher precision.

7.2 Risks and Mitigation

Risk	Severity	Mitigation
High latency (multiple Gemini calls)	High	Parallel tool calls, cached expansion
LLM hallucination	High	RAGAS faithfulness ≥ 0.87 , fallback templates
BM25 rebuild on restart	Low	Serialize BM25 index to disk
PostgreSQL cold start	Low	Connection pooling
Gemini API rate limit	Medium	Batch processing, exponential backoff

Table 7.1: Risk matrix and corresponding mitigation strategies

Chapter 8

CONCLUSION

8.1 Conclusion

This project successfully built an intelligent multimodal fashion search system following a clear two-phase roadmap. The RAG v1.0 phase established a hybrid search foundation with FashionCLIP 2.0 and BM25, while Agent v2.0 PATH 1 upgraded to a proactive architecture featuring a ReAct loop, intent classification, a clarification gate, and persistent memory via PostgreSQL.

Version v2.0 represents significant improvements over v1.0:

- **Embedding:** Upgraded from FashionCLIP (512-d) to Marqo-FashionSigLIP (768-d).
- **Query Expansion:** Added Gemini-powered synonym expansion for short queries.
- **Reranker:** Integrated BGE Reranker v2-m3 cross-encoder for improved precision.
- **Deployment:** Docker Compose stack with 4 services (PostgreSQL + Qdrant + FastAPI + Cloudflare Tunnel).

8.1.1 Evidence-Based Claim Mapping for Final Conclusion

Final claims are mapped to evidence strength so that conclusions remain proportional to verified data:

Claim	Primary Evidence	Status
Agent v2.0 improves retrieval quality over v1.0.	Recall@5, MRR, Hit Rate@5 (Section 6.2 and Section 6.2.2).	Supported
Agent behavior is robust in multi-turn interaction.	Task completion, faithfulness, and clarification behavior logs.	Supported
Operational LLM cost is explainable and monitorable.	llm_token_usage, session_token_summary, mode cost aggregation.	Partially supported
User purchase tendency is improved by recommendations.	CartAddRate / WillBuyRate / ConversionRate from behavior funnel.	Inconclusive (depends on cohort size and balance)

Table 8.1: Claim-to-evidence mapping used in the final thesis conclusion

Therefore, the final conclusion follows three rules: (1) only claims marked Supported are stated as confirmed findings, (2) Partially supported claims are reported with operational caveats, and (3) Inconclusive claims are retained as hypotheses for future data collection.

Current evidence limitations.

- Token-to-cost estimates depend on fixed public list prices and may vary over time.
- Behavior metrics are sensitive to sample imbalance across orchestration modes and user groups.
- Conversion-style claims require larger longitudinal cohorts than early-stage thesis collection windows.

Future development directions:

- Phase 3: Scale dataset to 15K–100K products, fine-tune the reranker, Redis session cache.
- Phase 4: Virtual Try-on (IDM-VTON), fine-tune FashionSigLIP on proprietary data, A/B testing.
- PATH 2: Image-to-Image search with Composed Search.

References

Appendix A

LISTINGS

A.1 Supplementary Equations and Derivations

This appendix contains supplementary mathematical material referenced in the main text.

A.2 Cosine Similarity from L2-Normalized Vectors

For two vectors $u, v \in \mathbb{R}^d$, the cosine similarity is:

$$\cos(u, v) = \frac{u \cdot v}{\|u\|_2 \|v\|_2} \quad (\text{A.1})$$

When both vectors are L2-normalized (i.e. $\|u\|_2 = \|v\|_2 = 1$), Equation (A.1) reduces to the dot product:

$$\cos(\hat{u}, \hat{v}) = \hat{u} \cdot \hat{v} \quad (\text{A.2})$$

This is why **FashionEmbedder** normalizes embeddings before storing them in Qdrant — it allows the use of fast dot-product search rather than full cosine computation.

A.3 BM25 Scoring Function

The BM25 score of a document d for a query $q = \{q_1, q_2, \dots, q_n\}$ is:

$$\text{BM25}(d, q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (\text{A.3})$$

where $f(q_i, d)$ is the term frequency of q_i in d , $|d|$ is the document length, avgdl is the average document length in the corpus, and k_1, b are free parameters (typically $k_1 \in [1.2, 2.0]$ and $b = 0.75$).

A.4 Sample Configuration Files

A.5 PostgreSQL Initialization Script

This appendix contains a sample SQL initialization script that may be used to bootstrap a fresh PostgreSQL instance for the Fashion Agent. See Table A.1 for the required environment variables.

Variable	Description
PGHOST	PostgreSQL host (e.g., <code>postgres</code> in Docker network)
PGPORT	PostgreSQL port (default: 5432)
PGDATABASE	Database name (<code>fashion_rag</code>)
PGUSER	Database user (<code>fashion_user</code>)
PGPASSWORD	Database password
GEMINI_API_KEY	Google Gemini API key
QDRANT_HOST	Qdrant host (e.g., <code>qdrant</code>)

Table A.1: Required environment variables for deployment